

Sa'ood Williams

FULL-STACK DEVELOPER

Cape Town, South Africa | saoodwilliams321@gmail.com

Portfolio and live demos | [GitHub: CACTUSCASH](#) | [LinkedIn](#)

PROFILE

Full-stack developer working with React, TypeScript, Next.js, Node.js and SQL. Founder of Cipher30, with responsibility for development, deployment and support. Independent project work covers collaborative editing, workflow execution, distributed-systems simulations and GPU graphics. Open to remote development roles and contracts.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, HTML, CSS

Applications: React, Next.js, Node.js, REST APIs, Server-Sent Events

Data and graphics: PostgreSQL, Supabase, SQLite, Row Level Security, WebGL2, GLSL, SVG

Delivery and testing: Git, GitHub Actions, Docker, Vercel, Node test runner, Vitest

EXPERIENCE

Founder & Full-Stack Developer | Cipher30

Self-employed | January 2024 to present | Cape Town / Remote

- Develop and operate an online education platform using Next.js, React and TypeScript; manage deployments, hosting, domains, DNS and first-line support.
- Resolved iOS Safari scroll performance problems by removing conflicting scroll behaviour, scheduling carousel updates with requestAnimationFrame and isolating frequent clock updates.
- Built a Node.js and discord.js lead-capture bot backed by SQLite, with source attribution, pipeline tracking and paid-member role commands.

cipher30mentorship.com

SELECTED INDEPENDENT PROJECTS | 2026

Canvas Rooms | JavaScript / Node.js / SQLite

github.com/CACTUSCASH/canvas-rooms

- Built a collaborative SVG diagram editor with shape editing, local persistence, JSON and SVG export, and real browser-tab presence.
- Implemented deterministic operation merging and deletion tombstones, with an optional SQLite-backed room service using Server-Sent Events and HTTP writes.

Quorum Lab | JavaScript / Distributed systems

github.com/CACTUSCASH/quorum-lab

- Built a deterministic five-node consensus simulator with elections, log replication, majority commits, network partitions and crash recovery.
- Tested log safety, stale-leader behaviour and recovery across seeded fault schedules. This is an educational simulator, not a production cluster.

Sa'ood Williams | Project work

Relay Workflows | Node.js / SQLite

github.com/CACTUSCASH/relay-workflows

- Implemented a shared workflow execution engine with validated graphs, typed transforms, conditional branches, payload traces and retry fixtures.
- Persisted workflow and execution history in SQLite and verified recovery after server restarts. Hosted demo runs locally in the browser; HTTP steps are labelled simulations.

Spectra Studio | WebGL2 / GLSL

github.com/CACTUSCASH/spectra-studio

- Built a live fragment-shader editor with five presets, portable saved patches and PNG export.
- Kept the previous shader active after compilation errors, cached uniforms and GPU resources, and stopped rendering when the canvas is hidden.

Atlas Pathfinder | Algorithms / Web Workers

github.com/CACTUSCASH/atlas-pathfinder

- Implemented A*, Dijkstra and breadth-first search on editable weighted maps with comparison, pause and single-step playback.
- Checked shortest-path results against an independent reference across 120 seeded maps; tested maze connectivity, worker execution and malformed imports.

Freelance Ledger | Next.js / TypeScript / Supabase

github.com/CACTUSCASH/freelance-ledger

- Built invoicing with email-link authentication, account-scoped PostgreSQL access rules and money calculations in integer minor units.
- Integrated currency conversion with timeout and fallback handling; included Docker configuration and CI checks in the repository.

Further project work

Orbit Workspace provides task editing, drag-and-drop, filtering and local persistence. Prism Analytics validates CSV imports, calculates revenue totals and exports filtered records. Both include local Node.js / SQLite backends. Additional repositories cover HTTP security-header reporting, MCP tools and GPU particle graphics.

EDUCATION

Higher Certificate in Information Systems (Cyber Security)

Eduvos | Completed

PROJECT DEMOS AND SOURCE

cactuscash.github.io/saoodwilliams-com

Hosted demonstrations run in the browser. Node.js / SQLite services are available in the source repositories for local use. Documentation describes setup, tests and project limitations.